

Chemical Bonds and Properties of Chemical Compounds

Why does salt dissolve in water but wax does not? Why does sugar melt at a relatively low temperature while salt requires intense heat to melt? Why can a copper wire conduct electricity but a plastic handle cannot? The answers lie in the chemical bonds that hold atoms together. The type of bond between atoms determines nearly every physical property of a compound – its melting point, boiling point, solubility, and electrical conductivity. Yet many students learn to name compounds without understanding why they behave the way they do.

The chemical formula of a compound is a symbolic representation of its chemical composition, telling you which elements are present and in what ratio. But the formula alone does not tell you everything. The properties of a compound depend on the type of bond between the elements. Ionic bonds, covalent bonds, and the polarity of molecules explain why substances as different as table salt and sugar, water and oil, metal and plastic behave so differently.

This reading material explores the differences between ionic and covalent bonds in terms of electrons involved, the physical properties of ionic and covalent compounds (using sodium chloride and water as examples), how to draw Lewis structures applying the octet rule, and how to predict molecular polarity based on electronegativity and molecular geometry.

Part 1: What Are Chemical Bonds and Why Do They Matter?

Definition of Chemical Bonds

A **chemical bond** is the force that holds atoms together in a compound. Chemical bonds form when atoms share or transfer electrons to achieve a stable electron configuration (usually a full outer energy level of 8 electrons, known as the octet rule).

The Central Idea: Bonds Determine Properties

The chemical formula of a compound tells you which elements are present and in what ratio (e.g., NaCl for sodium chloride, H₂O for water). However, the properties of a chemical compound depend on the type of bond between the elements. Ionic bonds produce different physical properties than covalent bonds.

Why Chemical Bonds and Properties Matter

Aspect	Explanation
Predicting behavior	Knowing the bond type helps predict solubility, melting point, and conductivity
Material selection	Engineers choose materials based on bond properties (e.g., metals for wiring, plastics for insulation)
Drug design	Pharmaceutical chemists consider bond types and polarity when designing medicines
Environmental science	Understanding solubility (polar vs. nonpolar) explains how pollutants move through water and soil
Everyday life	Why salt dissolves in water, why oil floats, why ice melts – all explained by bonds and polarity

Part 2: Ionic Bonds vs. Covalent Bonds – Differences in Electrons

Before exploring properties, you must understand the two main types of chemical bonds: ionic bonds and covalent bonds. The key difference lies in what happens to the electrons.

Ionic Bonds – Transfer of Electrons

An **ionic bond** forms when one atom transfers one or more electrons to another atom. This creates ions (charged atoms): one positive (cation) and one negative (anion). The opposite charges attract, forming the bond.

Feature	Explanation
Electron movement	Electrons are completely transferred from one atom to another
Resulting particles	Ions (cations and anions)
Types of elements	Typically metal + nonmetal
Example	Sodium chloride (NaCl): Na loses 1 electron $\rightarrow$ Na^+ ; Cl gains 1 electron $\rightarrow$ Cl^-
Bond strength	Strong (high melting and boiling points)

Covalent Bonds – Sharing of Electrons

A **covalent bond** forms when two atoms share one or more pairs of electrons. Neither atom completely loses or gains electrons; instead, they share electrons to achieve full outer shells.

Feature	Explanation
Electron movement	Electrons are shared between atoms
Resulting particles	Molecules (neutral groups of atoms)
Types of elements	Typically nonmetal + nonmetal

Feature	Explanation
Example	Water (H ₂ O): Each hydrogen shares one electron with oxygen; oxygen shares two electrons
Bond strength	Varies (single < double < triple bonds)

Comparison Table: Ionic vs. Covalent Bonds

Feature	Ionic Bond	Covalent Bond
Electron behavior	Transfer of electrons	Sharing of electrons
Types of elements	Metal + nonmetal	Nonmetal + nonmetal
Resulting structure	Crystal lattice (3D array of ions)	Discrete molecules or network covalent
Example	NaCl (sodium chloride)	H ₂ O (water), CO ₂ (carbon dioxide)
Melting point	High (often > 800°C)	Low to moderate
Solubility in water	Often soluble	Varies (polar soluble, nonpolar insoluble)
Electrical conductivity	Conducts when melted or dissolved	Does not conduct (except acids)

Part 3: The Octet Rule and Lewis Structures

The **octet rule** states that atoms tend to gain, lose, or share electrons to achieve a stable electron configuration with 8 electrons in their outermost energy level (like noble gases). Hydrogen is an exception, achieving stability with 2 electrons (like helium).

Drawing Lewis Structures for Ionic Compounds

A **Lewis structure** uses dots to represent valence electrons. For ionic compounds, show the transfer of electrons and the resulting ions with charges.

Example: Sodium Chloride (NaCl)

Step	Explanation	Result
1	Determine valence electrons	Na: 1 valence electron; Cl: 7 valence electrons
2	Identify electron transfer	Na loses 1 electron (becomes Na^+); Cl gains 1 electron (becomes Cl^-)
3	Draw the Lewis structure	$\text{Na}^+ [\text{Cl}]^-$ with 8 dots around Cl (full octet)

Lewis structure representation:

- Sodium atom: $\text{Na}\cdot$ (one dot for one valence electron)
- Chlorine atom: $\cdot\cdot\text{Cl}\cdot\cdot\cdot$ (seven dots for seven valence electrons)
- After transfer: Na^+ and $[\cdot\cdot\text{Cl}\cdot\cdot\cdot]^-$ (now with eight electrons around Cl)

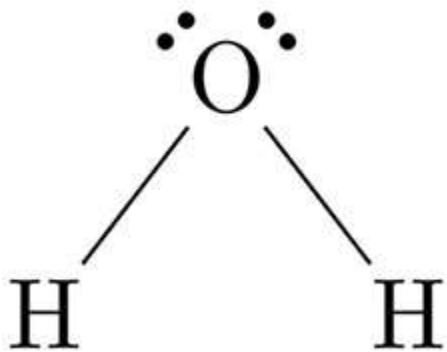
Drawing Lewis Structures for Covalent Compounds

For covalent compounds, show shared electron pairs as lines (bonding pairs) and lone pairs as dots.

Example: Water (H₂O)

Step	Explanation	Result
1	Determine total valence electrons	H: 1 each $\times$ 2 = 2; O: 6; total = 8 electrons
2	Determine central atom	Oxygen (least electronegative except hydrogen)
3	Place bonding pairs	Each H shares one electron with O (two single bonds)
4	Place remaining electrons as lone pairs	O has 4 remaining electrons $\rightarrow$ 2 lone pairs
5	Check octets	O has 8 electrons (2 bonds + 4 lone); each H has 2 electrons

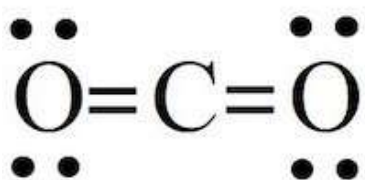
Lewis structure of water:



Example: Carbon Dioxide (CO₂)

Step	Explanation	Result
1	Total valence electrons	C: 4; O: 6 each × 2 = 12; total = 16 electrons
2	Central atom	Carbon
3	Place bonding pairs	Double bonds between C and each O to satisfy octets
4	Check octets	Each atom has 8 electrons

Lewis structure of carbon dioxide:



Steps for Drawing Lewis Structures (Summary)

Step	Action
1	Count total valence electrons (add for all atoms)
2	Determine the central atom (least electronegative, usually not hydrogen)
3	Connect atoms with single bonds (each bond = 2 electrons)
4	Distribute remaining electrons as lone pairs to complete octets
5	If the central atom lacks an octet, form double or triple bonds

Part 4: Physical Properties of Ionic vs. Covalent Compounds

The type of bond directly affects the physical properties of a compound. Using secondary sources (such as chemistry textbooks, scientific databases, and educational websites), you can create a table comparing the properties of an ionic compound (sodium chloride, NaCl) and a covalent compound (water, H₂O).

Comparison Table: Sodium Chloride (Ionic) vs. Water (Covalent)

Property	Sodium Chloride (NaCl) – Ionic	Water (H ₂ O) – Covalent
Chemical formula	NaCl	H ₂ O
Type of bond	Ionic (transfer of electrons)	Covalent (sharing of electrons)
Physical state at room temperature	Solid (crystalline)	Liquid
Melting point	801°C (very high)	0°C (low)
Boiling point	1,413°C (very high)	100°C (moderate)
Solubility in water	Highly soluble (37 g/100 mL)	Miscible (dissolves in itself)
Solubility in nonpolar solvents (e.g., oil)	Insoluble	Insoluble (water is polar; oil is nonpolar)
Electrical conductivity (solid)	Does not conduct (ions fixed in lattice)	Does not conduct (no free ions or electrons)
Electrical conductivity (melted or dissolved)	Conducts electricity (mobile ions)	Does not conduct (pure water has no ions)
Hardness	Hard but brittle	Liquid (not applicable)

Explanation of Differences

Property	Why Ionic Compounds (NaCl) Behave This Way	Why Covalent Compounds (H ₂ O) Behave This Way
High melting/boiling points	Strong electrostatic forces between ions require much energy to overcome	Weak intermolecular forces (hydrogen bonds) require less energy to overcome
Solubility in water	Polar water molecules attract and separate ions (like dissolves like)	Water molecules are polar; they attract each other and other polar substances
Electrical conductivity	Ions are mobile when melted or dissolved, allowing charge to flow	No free ions or electrons; molecules are neutral
Hardness (NaCl)	Ions in crystal lattice resist separation but can shatter under force	Not applicable to water (liquid)

Part 5: Polarity of Compounds – Electronegativity and Molecular Geometry

Polarity describes how evenly electrons are distributed in a molecule. A polar molecule has a partial positive end and a partial negative end (a dipole). A nonpolar molecule has an even distribution of charge.

Electronegativity

Electronegativity is the ability of an atom to attract shared electrons in a chemical bond. The greater the difference in electronegativity between two bonded atoms, the more polar the bond.

Electronegativity Difference	Bond Type	Example
0.0 – 0.4	Nonpolar covalent	H-H, C-H
0.5 – 1.7	Polar covalent	H-O, H-Cl
> 1.7	Ionic (or mostly ionic)	Na-Cl

Predicting Polarity of a Molecule

To predict whether a molecule is polar or nonpolar, you must consider two factors:

Factor	Explanation
Electronegativity difference	Are the bonds themselves polar? (Atoms with different electronegativities create polar bonds)
Molecular geometry (shape)	Do the polar bonds cancel each other out, or do they add up to create a net dipole?

Common Molecular Geometries and Polarity

Molecular Shape	Example	Bond Polarity	Symmetry	Overall Polarity
Linear (symmetric)	CO ₂	Polar bonds (C=O)	Symmetric	Nonpolar (bonds cancel)
Linear (asymmetric)	HCl	Polar bond (H-Cl)	Asymmetric	Polar
Bent (V-shaped)	H ₂ O	Polar bonds (O-H)	Asymmetric	Polar
Trigonal planar	BF ₃	Polar bonds (B-F)	Symmetric	Nonpolar (bonds cancel)

Molecular Shape	Example	Bond Polarity	Symmetry	Overall Polarity
(symmetric)		F)		cancel)
Trigonal pyramidal	NH ₃	Polar bonds (N-H)	Asymmetric	Polar
Tetrahedral (symmetric)	CH ₄	Nonpolar bonds (C-H)	Symmetric	Nonpolar

Examples of Polarity Prediction

Example 1: Carbon Dioxide (CO₂)

Factor	Analysis
Electronegativity difference	C (2.55) vs. O (3.44): difference = 0.89 → polar bonds
Molecular geometry	Linear (O=C=O)
Polarity	Polar bonds are equal and opposite → cancel each other → nonpolar molecule

Example 2: Water (H₂O)

Factor	Analysis
Electronegativity difference	H (2.20) vs. O (3.44): difference = 1.24 → polar bonds
Molecular geometry	Bent (V-shaped) due to two lone pairs on oxygen
Polarity	Polar bonds do not cancel; they add up to create a net dipole → polar molecule

Example 3: Methane (CH₄)

Factor	Analysis
Electronegativity difference	C (2.55) vs. H (2.20): difference = 0.35 → essentially nonpolar bonds
Molecular geometry	Tetrahedral (symmetric)
Polarity	Symmetric shape with nonpolar bonds → nonpolar molecule

Electronegativity Values of Common Elements (Periodic Table Reference)

Element	Electronegativity (Pauling scale)
Fluorine (F)	4.0 (highest)
Oxygen (O)	3.44
Chlorine (Cl)	3.16
Nitrogen (N)	3.04
Bromine (Br)	2.96
Carbon (C)	2.55
Sulfur (S)	2.58
Hydrogen (H)	2.20
Phosphorus (P)	2.19
Boron (B)	2.04
Metals (Na, Mg, etc.)	0.7 – 1.5

Summary: "Like Dissolves Like"

The principle of "**like dissolves like**" explains solubility: polar solvents (like water) dissolve polar solutes and ionic compounds; nonpolar solvents (like oil, hexane) dissolve nonpolar solutes.

Solvent Polarity	Dissolves	Does Not Dissolve
Polar (e.g., water)	Polar molecules, ionic compounds	Nonpolar molecules (oil, wax)
Nonpolar (e.g., oil, hexane)	Nonpolar molecules (fats, waxes)	Polar molecules, ionic compounds

Key Terms

Term	Definition
Chemical bond	The force that holds atoms together in a compound
Ionic bond	A bond formed by the complete transfer of electrons from one atom to another
Covalent bond	A bond formed by the sharing of electrons between atoms
Ion	A charged atom (cation = positive, anion = negative)
Molecule	A neutral group of atoms held together by covalent bonds
Octet rule	Atoms tend to gain, lose, or share electrons to achieve 8 valence electrons
Lewis structure	A diagram showing valence electrons as dots and bonds as lines
Electronegativity	The ability of an atom to attract shared electrons in a bond
Polarity	The uneven distribution of charge in a molecule
Dipole	A molecule with a positive end and a negative end
Nonpolar	A molecule with even charge distribution
Like dissolves like	A rule stating that polar solvents dissolve polar solutes, and nonpolar solvents dissolve nonpolar solutes

Review Questions

1. Identify the two main types of chemical bonds. Explain the difference between them in terms of what happens to the electrons.
2. What is the octet rule? Explain why atoms form chemical bonds based on this rule.
3. Draw the Lewis structure for the ionic compound sodium chloride (NaCl). Show the electron transfer and the resulting ions with charges.
4. Draw the Lewis structure for the covalent compound water (H₂O). Show all bonding pairs and lone pairs. Verify that the octet rule is satisfied for oxygen.
5. Using secondary sources (or the information in this lesson), complete the following table comparing sodium chloride (ionic) and water (covalent):

Property	Sodium Chloride (NaCl)	Water (H ₂ O)
Physical state at room temperature		
Melting point		
Solubility in water		
Electrical conductivity (solid)		
Electrical conductivity (melted or dissolved)		

6. Explain why sodium chloride has a very high melting point (801°C) while water has a low melting point (0°C). Use the terms ionic bond, covalent bond, and intermolecular forces in your answer.
7. Define electronegativity. How does the difference in electronegativity between two atoms determine whether a bond is nonpolar covalent, polar covalent, or ionic?

8. Predict whether each of the following molecules is polar or nonpolar. Justify your answer using electronegativity difference and molecular geometry.

- (a) Carbon dioxide (CO_2)
- (b) Water (H_2O)
- (c) Methane (CH_4)
- (d) Ammonia (NH_3)

9. Explain the rule "like dissolves like." Use this rule to predict whether sodium chloride (NaCl) will dissolve in water (polar) and whether oil (nonpolar) will dissolve in water.

10. Application question: A student finds two unlabeled white powders in the laboratory. One is sodium chloride (NaCl , ionic) and the other is sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$, covalent). Describe two simple physical tests the student could perform to identify which powder is which. For each test, explain what result would indicate ionic and what result would indicate covalent.